

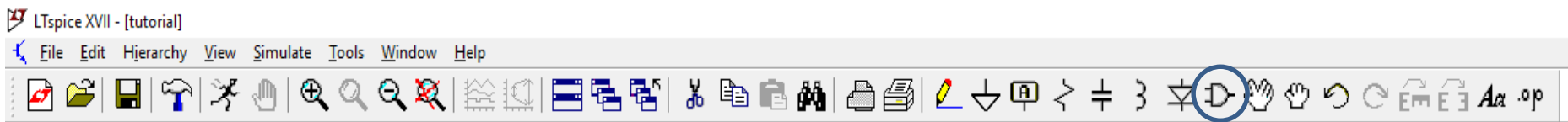
LTspiceTutorial

Getting Started

- Download LTSPICE IV from <http://www.linear.com/designtools/software/>
- Download the 0.18um library file '**log018.l**' from Brightspace, & copy it to **C:\Program Files\LTC\LTspiceXVII\lib\cmp**
- Open the .l library file in LTSPICE & change the library paths in all the .lib definitions to -
C:\Program Files\LTC\LTspiceXVII\lib\cmp\log018.l

LTSPICE: Schematic

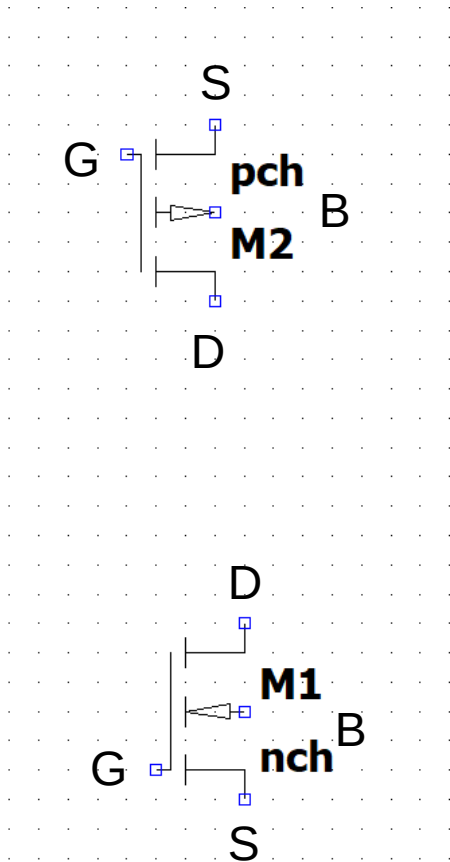
- Go to File->New Schematic
- Include the library file by using a SPICE directive –
 .lib 'C:\Program Files\LTC\LTspiceXVII\lib\cmp\log018.l' TT
- Go to Edit->Component and insert the following components in the schematic-
 nmos4 – NMOS
 pmos4 – PMOS
 voltage – Voltage source



LTSPICE: Schematic

- Right click on NMOS & PMOS and edit the model name from NMOS/PMOS to nch and pch, respectively
- Enter relevant parameters for transistors(W,L and M; ignore AD, AS, etc for now) and voltage sources
- Go to Edit->Draw Wire and connect all the components
- Go to Edit->Place GND and place the ground in the circuit
- Go to Edit->Label Net and label all the net names you might want to plot later (makes life easier)
- To name a voltage source, right click on its name and edit
- Go to Simulate->Control Panel->Netlist options, and uncheck Default models and default libraries

Transistors in Schematic



Monolithic MOSFET - M2

Model Name:	pch	OK	Cancel
Length(L):	180n		
Width(W):	10u		
Drain Area(AD):			
Source Area(AS):			
Drain Perimeter(PD):			
Source Perimeter(PS):			
No. Parallel Devices(M):	2		

pch l=180n w=10u m=2

Monolithic MOSFET - M1

Model Name:	nch	OK	Cancel
Length(L):	180n		
Width(W):	10u		
Drain Area(AD):			
Source Area(AS):			
Drain Perimeter(PD):			
Source Perimeter(PS):			
No. Parallel Devices(M):	2		

nch l=180n w=10u m=2

- Transistors are not symmetrical.
- The terminal which is closer to the gate of the transistor is the source terminal.
- The body of pmos (pch) can be connected either to the source of the transistor or the most positive voltage (VDD)
- The body of nmos (nch) must be connected to the most negative voltage (GND or VEE)

LTSPICE: DC analysis

- Go to Edit->SPICE Analysis->DC sweep, and enter the following –

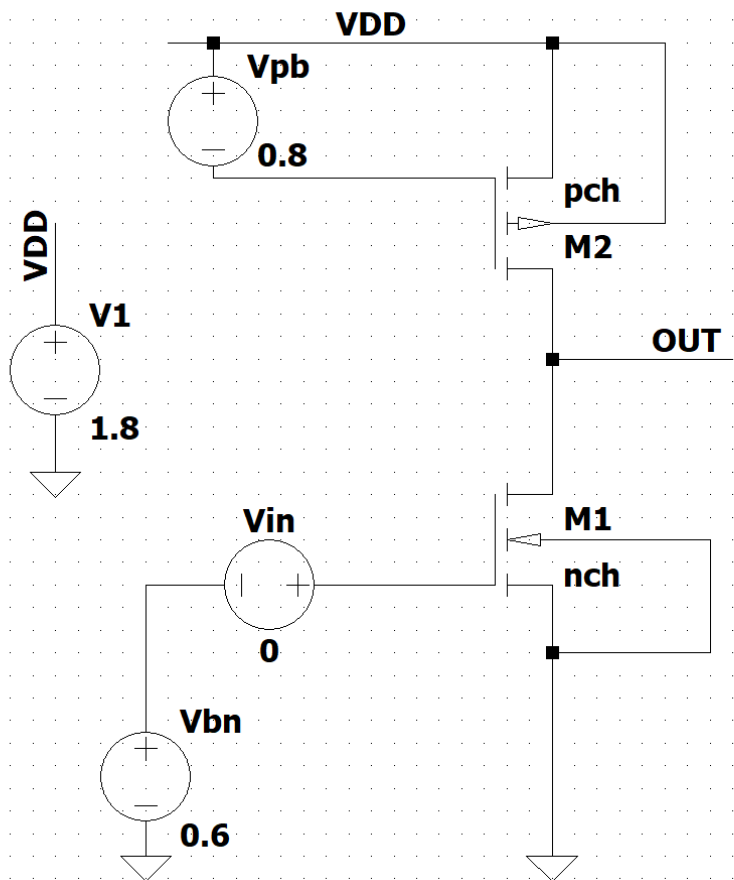
Name of 1st Source to Sweep: Vgsn

Type of Sweep: Linear

Start Value: -0.4

Stop Value: 1

Increment: 0.001



```
.lib 'C:\Program Files\LTC\LTspiceXVII\lib\cmp\log018.l' TT
.dc Vin -0.4 1 0.001
```

LTspice XVII - [tutorial]

File Edit Hierarchy View Simulate Tools Window Help



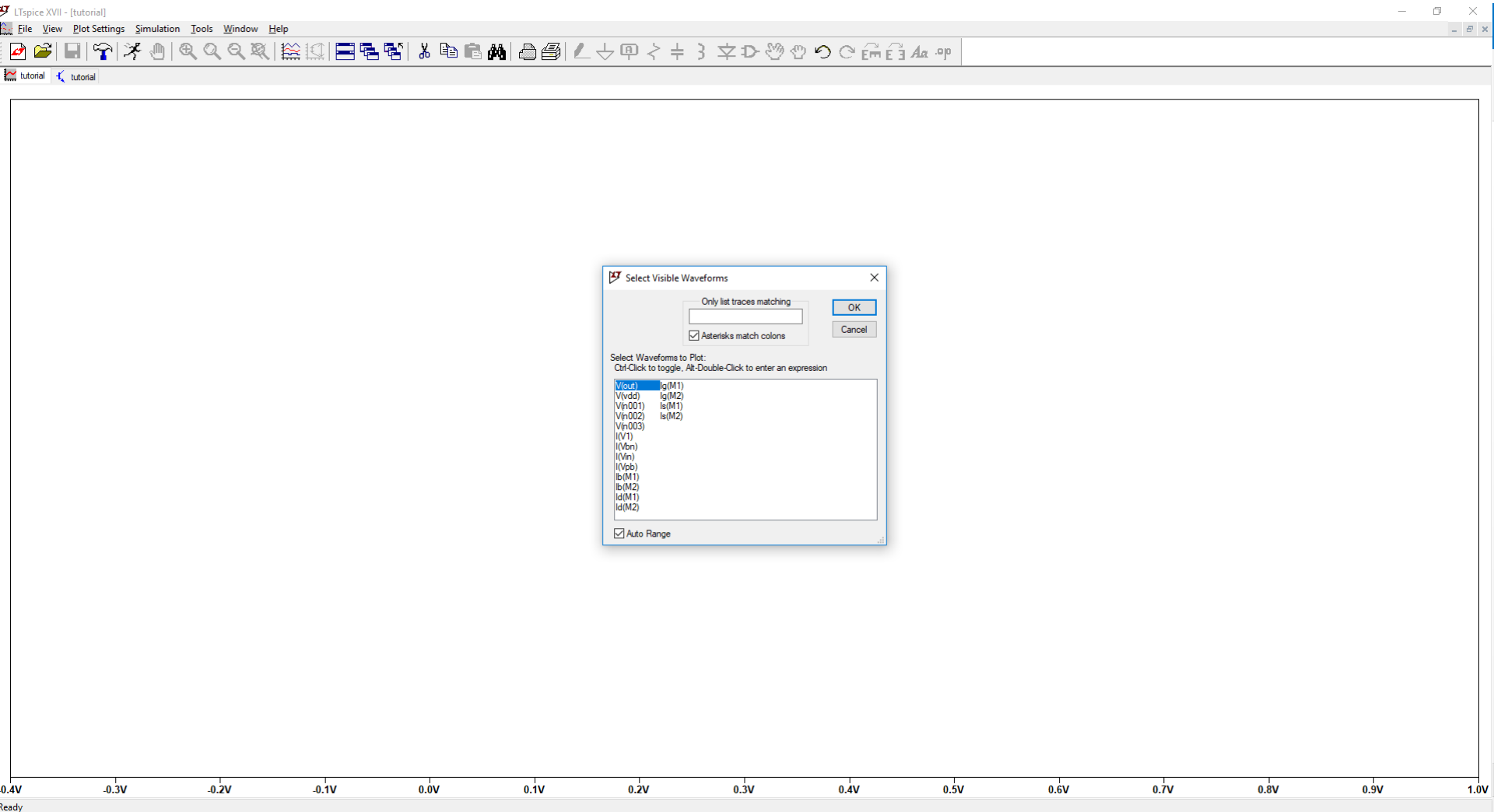
Press run

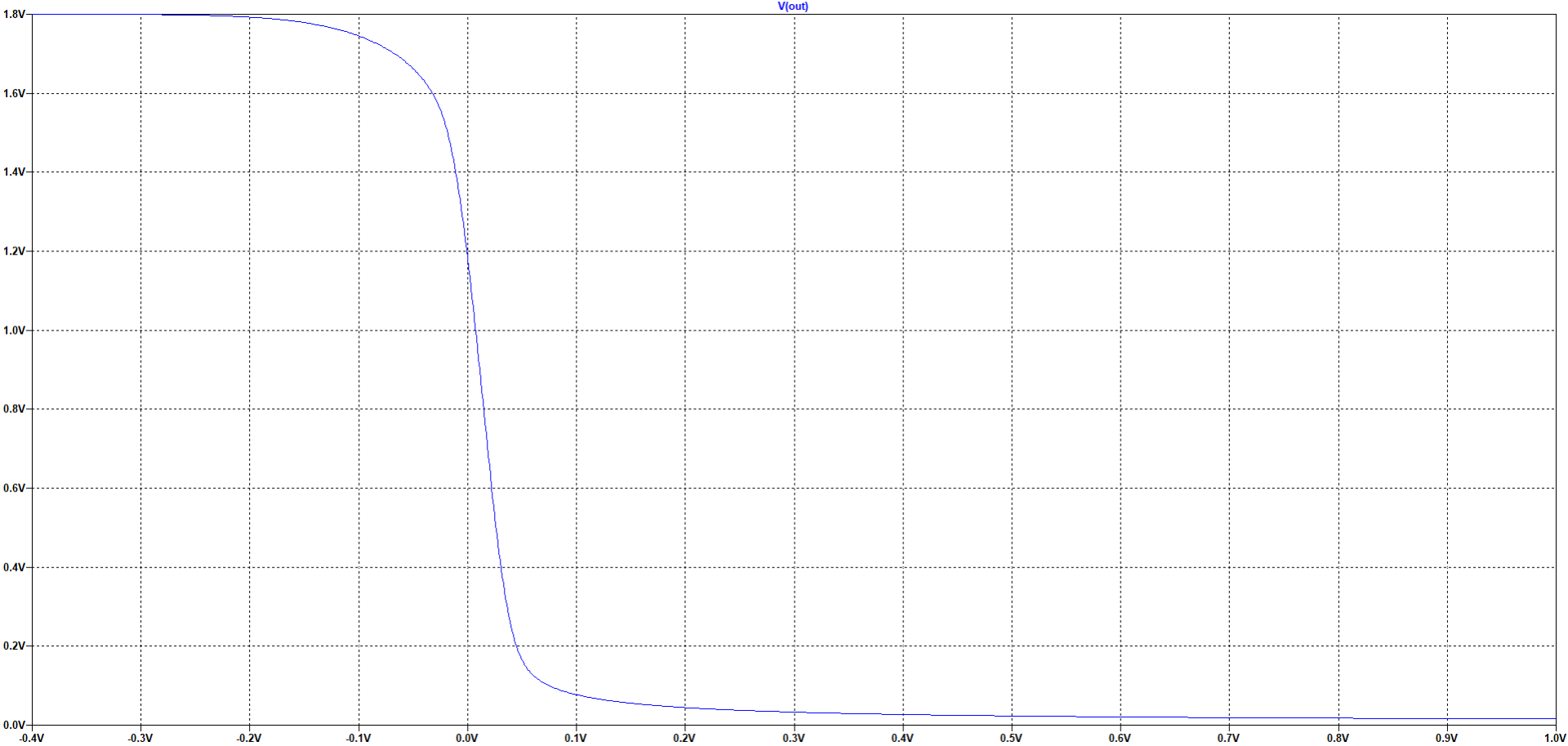
LTSPICE: Simulation Results

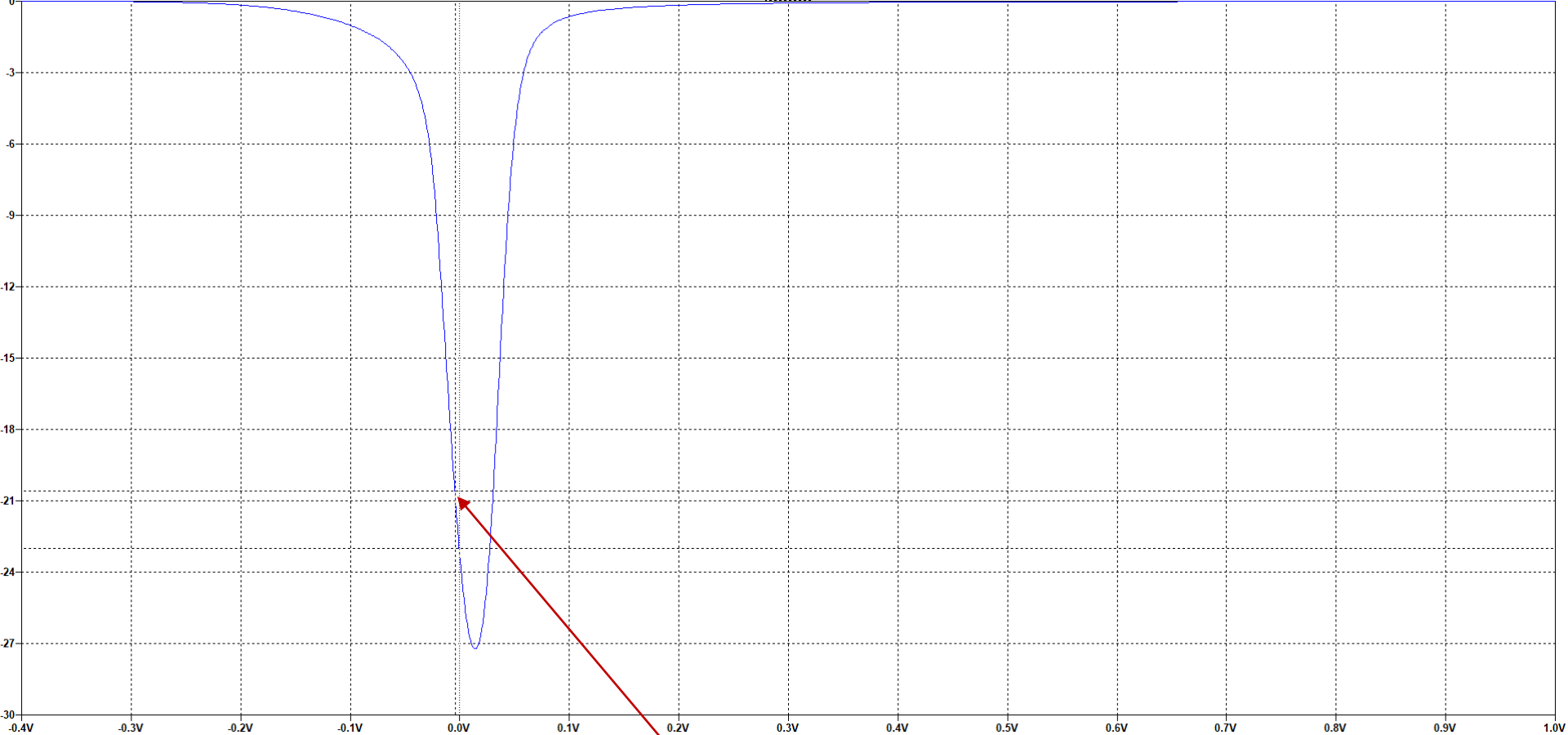
- Successful simulation opens a .raw file in another window
- All the plots are saved in this .raw file
- To plot, Right click on the plot area & select Visible Traces

LTSPICE: DC Simulation Results

- To plot gain, Right click on the plot area & select Add Trace
- Use $d()$ to calculate the derivative of the relevant curve w.r.t the swept variable
- For complete list of mathematical operations, go to Help->Help Topics->Waveform Arithmetic



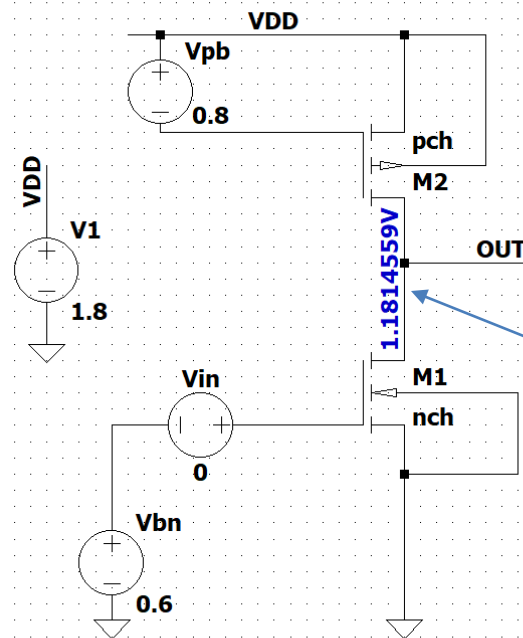




Gain is -23 when $V_{in_{pp}} \approx 0$

LTSPICE: DC op pnt analysis

- Calculates DC operating point of the circuit
- Go to Edit->SPICE Analysis->DC op pnt
- Successful simulation opens a list of all node voltages and currents in the circuit
- To annotate node voltages on the schematic, right click on the schematic and select View->Place .op Data Label and select the desired net
- To view the detailed operating point of the transistors, go to View->SPICE Error Log (Ctrl+L)



```
.lib 'C:\Program Files\LTC\LTspiceXVII\lib\cmp\log018.' TT
.dc Vin -0.4 1 0.001
.op
```

```

--- BSIM3 MOSFETS ---
Name:      m2      m1
Model:     pch.6   nch.6
Id:        -4.26e-04  4.26e-04
Vgs:       -8.00e-01  6.00e-01
Vds:       -6.19e-01  1.18e+00
Vbs:       0.00e+00   0.00e+00
Vth:       -4.99e-01  5.07e-01
Vdsat:     -2.53e-01  1.05e-01
Gm:        2.32e-03   4.75e-03
Gds:       1.11e-04   9.34e-05
Gmb        7.86e-04   1.32e-03
Cbd:       0.00e+00   0.00e+00
Cbs:       0.00e+00   0.00e+00
Cgsbv:     6.54e-15   7.32e-15
Cgdov:     6.54e-15   7.32e-15
Cgbv:      9.65e-17   6.37e-17
dQgdVgb:   3.70e-14   3.62e-14
dQgdVdb:   -6.49e-15  -7.28e-15
dQgdVsb:   -2.91e-14  -2.53e-14
dQddVgb:   -6.60e-15  -7.32e-15
dQddVdb:   6.59e-15   7.32e-15
dQddVsb:   3.90e-17   9.00e-18
dQbdVgb:   -4.96e-15  -4.82e-15
dQbdVdb:   -7.05e-18  6.66e-18
dQbdVsb:   -2.77e-15  -3.55e-15

```

Date: Mon Nov 26 16:11:00 2018

LTSPICE: AC analysis

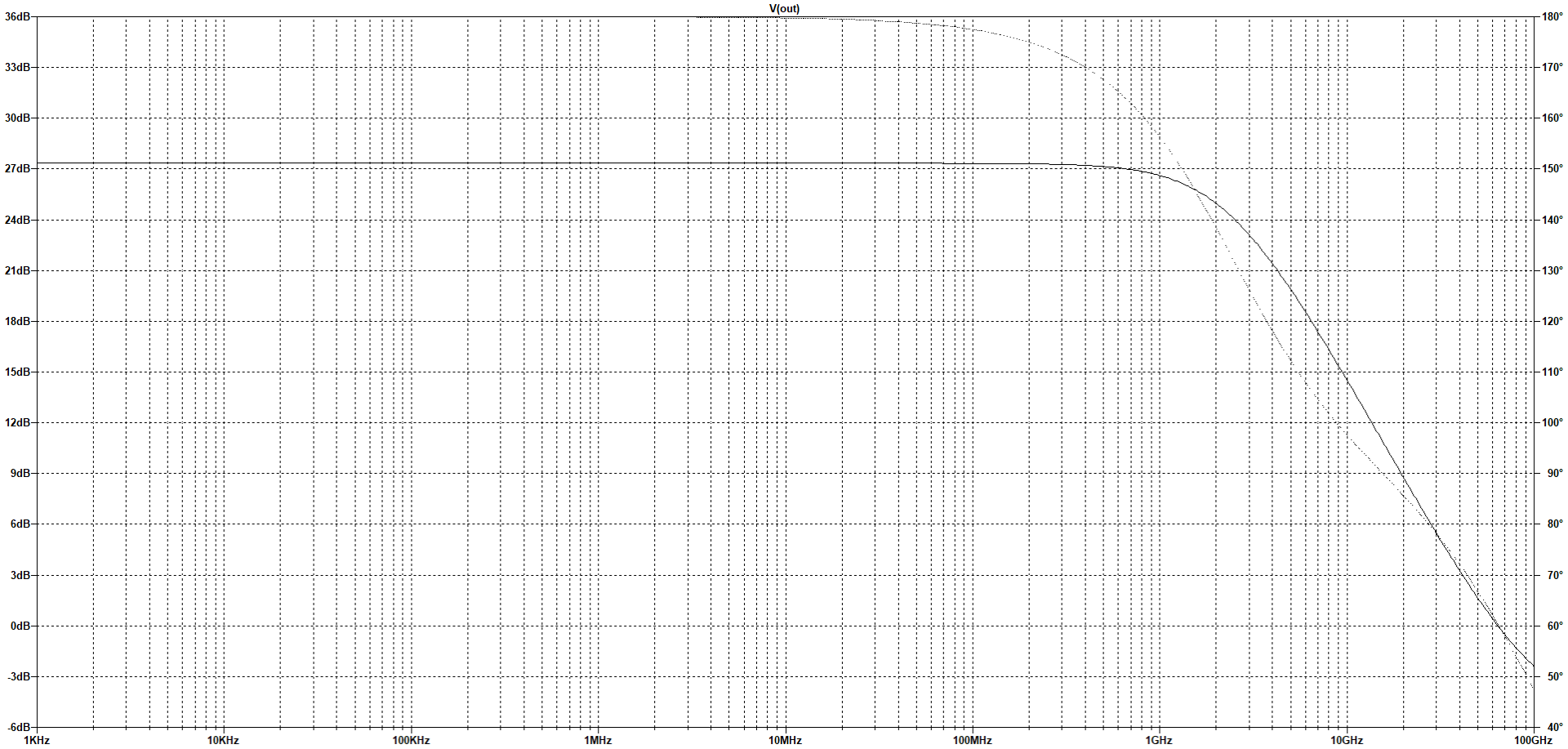
- Right click on Vin, Select Advanced and make AC amplitude=1
- Go to Edit->SPICE Analysis->AC analysis, and enter the following –

Type of Sweep: Decade

Number of points per decade: 20

Start Frequency: 1k

Stop Frequency: 100G



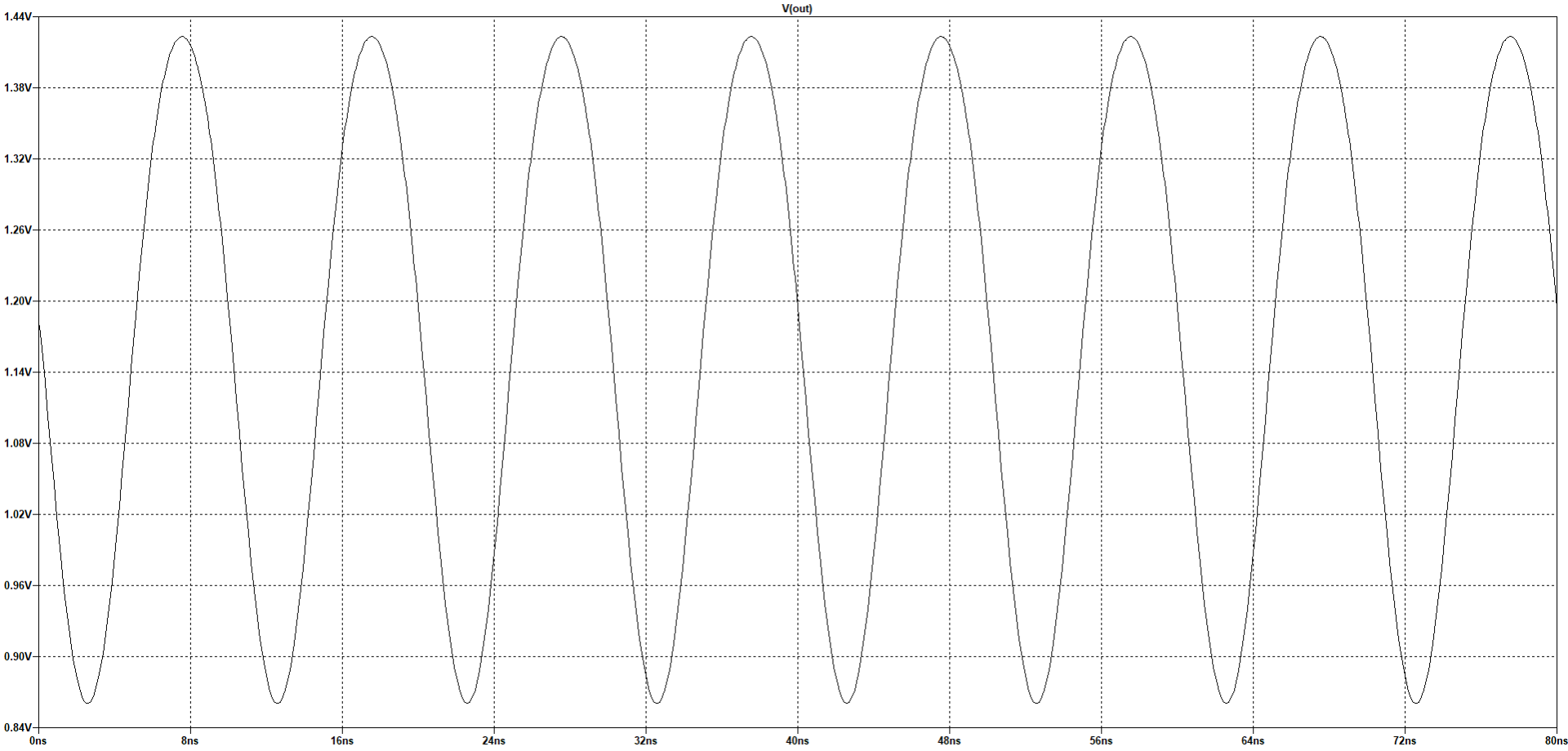
Ctrl+click on the $V(out)$

Waveform: $V(out)$ ✕

Start Frequency:	1KHz
End Frequency:	100GHz
Reference:	27.34dB @ 1KHz
BW:	10.911GHz
Power BW:	3.6703GHz

LTSPICE: Transient analysis

- Right click on Vin, Select Advanced, Select Sine under Functions & enter the following parameters –
DC Offset = 0, Amplitude = 12.5m, Freq = 100Meg
- Go to Edit->SPICE Analysis->Transient analysis, and enter the following –
Stop Time: 80n



LTSPICE: Noise analysis

- Go to Edit->SPICE Analysis->Noise, and enter the following :

Output: V(OUT)

Input: Vin

Type of Sweep: Decade

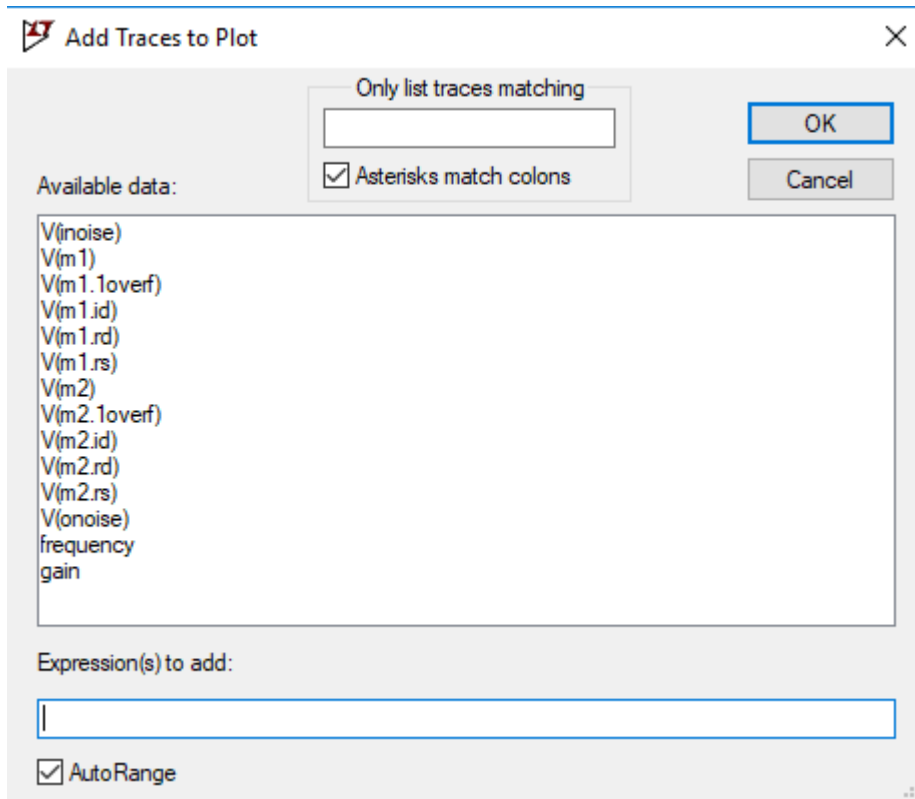
Number of points per decade: 20

Start Frequency: 1

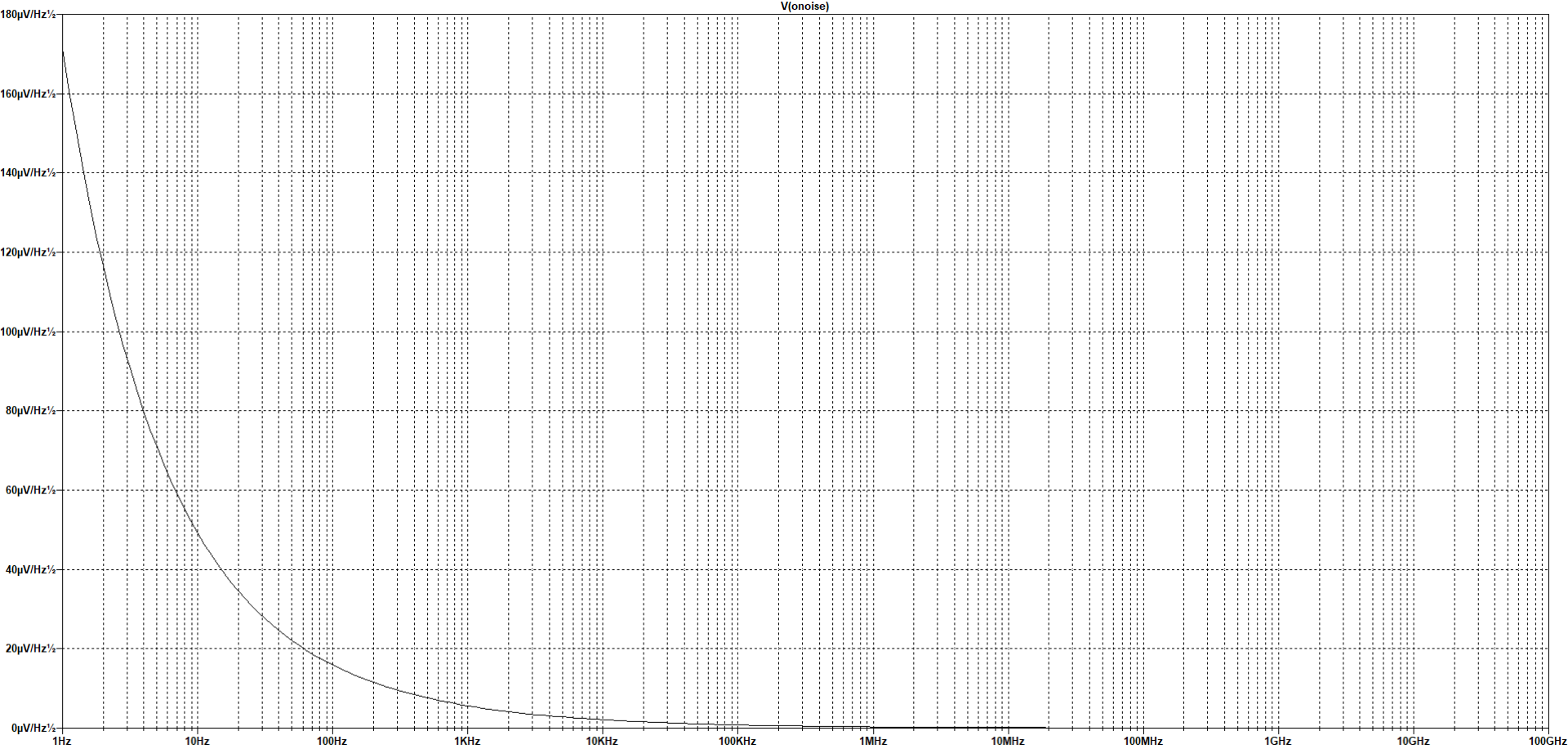
Stop Frequency: 100G

LTSPICE: Noise Analysis

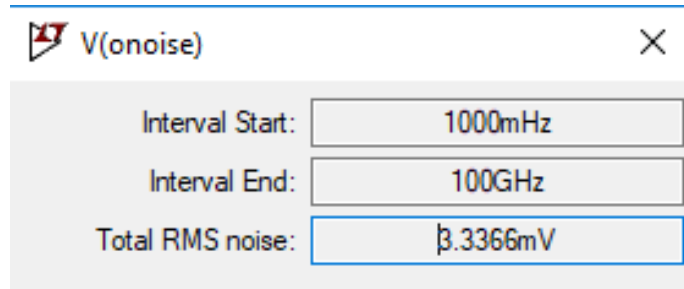
Right click on the plot area & select Add Trace



- Vnoise: Input referred noise
- Vonise: Output noise
- Individual contribution of each elements and noise types can be viewed clicking on the related plot



Ctrl+Click to V(noise) to observe the total rms noise



LTSPICE: Parametric analysis

- Used to run simulations for different values of a design variable
- Right click on Vin, Select Advanced, Select Sine under Functions & enter the following parameters –
DC Offset = 0, Amplitude = {Vin}, Freq = 100Meg
- Add the parameter definition by using the SPICE directive –
.param Vin 12.5m
- Sweep Vin by using the SPICE directive –
.step param Vin list 12.5m 25m 50m 100m
- For different kinds of parameter sweeps, go to Help Topics->.STEP

ref: <http://web.cecs.pdx.edu/~prasads/LTSpice%20Sweep%20Tutorial.pdf>

Misc.

- Shortcuts -

Draw wire F3

Component F2

Move F7

Label Net F4

SPICE Directive s

Place GND g

Copy F6 or Ctrl+C

Rotate F7->select object->Ctrl+R

Zoom to fit Space

Undo F9

Add resistor R

Add capacitor C

Add inductor L



Misc.

- To paste plots in word documents, Tools->Copy bitmap to clipboard

Further Reading

- **FFT set-up** : Basic Analog CMOS Design: An intuitive approach, by Klaas Bult
- **LTSPICE** :
 - Brief - http://www.ece.uidaho.edu/ee/classes/ECE211/pdf/LTSpice_Basic_Tutorial.pdf
 - Detailed - <http://ltspice.linear.com/software/scad3.pdf>
 - <http://denethor.wlu.ca/ltspice/>